MIL-F-62563(AT)
25 July 1988
SUPERSEDING
(see 6.4)

MILITARY SPECIFICATION

FAN, VENTILATING, VEHICULAR COMPARTMENT

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electric-powered, single-stage, vaneaxial fan, referred to herein as "fan", used to ventilate the crew compartment of an armored military vehicle (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4140
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SPECIFICATIONS
FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatment for Organic Coatings.

MILITARY

MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-C-22750	- Coating Epoxy - Polyamide.
MIL-P-23377	- Primer Coatings: Epoxy - Polyamide, Chemical and Solvent Resistant.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-P-53030	- Primer Coatings, Epoxy, Water Reducible, Lead and Chromate Free.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1275	- Characteristics of 28 Volt DC Electrical Systems in Military Vehicles.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Drawings. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING
ARMY

12292099	- Fan, Ventilating.

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(Copies of specifications, standards and drawings required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

AMCA-STD-210 - Laboratory Methods of Testing Fans for Rating Purposes.

(Application for copies should be addressed to the Air Moving and Conditioning Association, Inc., 30 West University Drive, Arlington Heights, Illinois 60004).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish fans which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent fans delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free from defects that could affect their serviceability or the performance of the finished product (see 4.7.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

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3.3 Design and construction. The fan shall be constructed in accordance with Drawing 12292099 and as specified herein (see 4.7.1 and 4.7.2).

3.3.1 Weight. The weight of the fan shall not exceed 12 pounds (lb) (see 4.7.3).

3.3.2 Electric motor. The motor shall be a series-wound, three-speed, variable, 24 volt (V) direct current (dc) type, directly coupled to the fan shaft. The motor shall be operable with electric power as specified in MIL-STD-1275 (see 4.7.1 and 4.7.2).

3.4 Performance.

3.4.1 Voltage. The fan shall operate with any control voltage from 11 to 30 V dc (see 4.7.4.1).

3.4.2 Operational performance. The fan shall show no evidence of an abnormal noise or vibration when 26 plus or minus (±) 2 V dc is applied on the electric motor. The direction of airflow shall be as specified on Drawing 12292099. The current shall not exceed 29.5 amperes (A) (see 4.7.4.2).

3.4.3 Rated performance. The fan shall deliver air at the rate specified in table I under the conditions specified therein, at 29.5 A or less (see 4.7.4.3).

TABLE I. Rated performance.

Minimum airflow rate (Standard cubic feet per minute)	Conditions	
	Input voltage (V dc) 1/	Minimum static pressure [Inches of water (gage)]
770	27	2.7
635	27	4.2
600	19.8	1.9
350	11.8	1.1

1/ Tolerance: +0.5 V dc, - 0.0 V dc.

3.4.4 Electromagnetic interference suppression. The fan shall meet the requirements of MIL-STD-461A, notice 4, limits RE05 and CE07, except that RE05 limits may be exceeded by no more than 20 decibels (dB) in the frequency ranges of 150 to 400 kilohertz (kHz) and 50 to 300 megahertz (MHz) (see 4.7.4.4).

3.4.5 Insulation resistance. The insulation resistance of the fan shall be not less than 12 megohms at 500 V dc at a relative humidity of 40 to 80 percent (%) (see 4.7.4.5).

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3.4.6 Dielectric strength. The fan shall withstand a voltage of 500 V root-mean-square at 60 hertz (Hz) for 1 minute with no evidence of arcing or breakdown (see 4.7.4.6).

3.4.7 Noise level. The sound power level of the fan when operating as specified in 3.4.3 shall not exceed the values shown in table II, when measured at a distance of 10 feet in any direction (see 4.7.4.7).

TABLE II. Sound power level.

Octave band (Hz)	Sound power level -12 (dB re 10 W) 1/
63	84
125	97
250	88
500	99
1000	96
2000	97
4000	92

1/ Reference (re), watt (W).

3.4.8 Environmental.

3.4.8.1 High temperature. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored at temperatures up to 160 degrees Fahrenheit (°F) and operated at temperatures up to 130°F (see 4.7.4.8.1).

3.4.8.2 Low temperature. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored at temperatures down to minus (-) 70°F and operated at temperatures down to -65°F (see 4.7.4.8.2).

3.4.8.3 Steam and waterjet cleaning. The fan shall show no evidence of damage or deterioration and demonstrate no performance degradation following a steam and waterjet cleaning process which uses a cleaner conforming to P-C-437, and which applies the jet perpendicular to the fan surface from a distance of not more than 1 foot for steam cleaning and not more than 5 feet for waterjet cleaning. The jet pressure shall be 105 $\pm$ 5 pound per square inch gage (psig) and the jet shall be moved at such a rate that the entire surface of the fan is subjected to the jet for a period of 1 minute (see 4.7.4.8.3).

3.4.8.4 Humidity. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after exposure to a relative humidity ranging from 0 to 98% (see 4.7.4.8.4).

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3.4.8.5 Shock. The fan shall show no evidence of damage or deformation and demonstrate no performance degradation after exposure to sawtooth shock pulses of 40 gravity units (g) for 11 milliseconds (ms) time duration applied along each of the three major axes (see 4.7.4.8.5).

3.4.8.6 Vibration. The fan shall show no evidence of damage or deformation and demonstrate no performance degradation during and after exposure to vibration levels of 1.5 and 4.2 g in the frequency range of 5 to 500 to 5 Hz with a sweep time of 15 minutes (see figure 1 and 4.7.4.8.6).

3.4.9 Durability. The fan shall be capable of operating 1000 hours continuously at 27 ± 2 V dc and 2 ± 0.2 inches of water pressure without performance degradation and without any maintenance or malfunction (except brushes may be replaced after not less than 500 hours). During the 1000 hours, the fan shall be in an ambient temperature of 70 ± 20°F and shall be switched off and on every 24 hours. A 1 minute off time shall be employed (see 4.7.4.9).

3.5 Finish. Exposed ferrous and aluminum surfaces shall be cleaned, treated, primed and finished as specified herein (see 4.7.2).

3.5.1 Ferrous metals.

3.5.1.1 Cleaning and treatment.

a. Clean ferrous metals (Rockwell hardness less than C40) in accordance with the applicable method (I through VI) specified in TT-C-490. Treat in accordance with type I or III of TT-C-490.

b. Clean ferrous metals (Rockwell hardness C40 through C47) in accordance with the applicable method (I through VI) specified in TT-C-490. Treat in accordance with type I or III of TT-C-490 (bake for hydrogen embrittlement relief as required).

c. Clean ferrous metals (Rockwell hardness C48 and over) in accordance with method I of TT-C-490. Acid solutions shall not be used for cleaning.

3.5.1.2 Priming. Apply a primer coating in accordance with MIL-P-53030, MIL-P-53022 or MIL-P-52192, composition optional. The dry film thickness of the primer shall be not less than 1.0 mil.

3.5.1.3 Finishing. Apply an epoxy-polyamide coating conforming to type I of MIL-C-22750 or a modified high solids epoxy which meet the performance requirements of MIL-C-22750. The dry film thickness of the coating shall be not less than 1.0 mil. The color shall match color chip number 24533 (green) of FED-STD-595.

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3.5.2 Aluminum.

3.5.2.1 Treatment. Treat in accordance with class 1A of MIL-C-5541.

3.5.2.2 Priming. Apply a primer coating in accordance with MIL-P-53030, MIL-P-53022 or MIL-P-23377, composition optional. The dry film thickness of the primer shall be not less than 1.0 mil.

3.5.2.3 Finishing. Apply an epoxy-polyamide coating conforming to type I of MIL-C-22750 or a modified high solids epoxy which meet the performance requirements of MIL-C-22750. The dry film thickness of the coating shall be not less than 1.0 mil. The color shall match color chip number 24533 (green) of FED-STD-595.

3.6 Identification marking. Nameplates and product marking shall be as specified on Drawing 12292099 (see 4.7.2).

3.7 Workmanship. The fan shall be free of defects such as burrs, sharp edges, corrosion, soil, dirt and any defects that would effect its performance or appearance (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Test (see 4.5.3).
c. Control test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18°F
b. Barometric pressure 28.5 + 2 inches of mercury
   - 3
c. Relative humidity 50 ± 30%

4.3.1 Environmental tests.

4.3.1.1 Test temperature stabilization. Test temperature stabilization is the point at which the test sample will not vary more than 3.6°F per hour.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select three fans produced under the production contract for first article inspection. First article samples shall be inspected as specified in table III, in the order specified in table IV. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply fans that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the fans of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following AQL's.

Classification	AQL
Major	1.0
Minor	2.5

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Test	
Materials and construction	3.2, 3.3 and 3.3.2	4.7.1	X			
Defects (see table V)	3.3, 3.3.2 3.5, 3.6 and 3.7	4.7.2	X	X		
Weight	3.3.1	4.7.3	X			
Voltage	3.4.1	4.7.4.1	X			X
Operational performance	3.4.2	4.7.4.2	X		X	X
Rated performance	3.4.3	4.7.4.3	X			X
Electromagnetic interference suppression	3.4.4	4.7.4.4	X			
Insulation resistance	3.4.5	4.7.4.5	X			X
Dielectric strength	3.4.6	4.7.4.6	X			X
Noise level	3.4.7	4.7.4.7	X			
High temperature	3.4.8.1	4.7.4.8.1	X			
Low temperature	3.4.8.2	4.7.4.8.2	X			
Steam and water-jet cleaning	3.4.8.3	4.7.4.8.3	X			
Humidity	3.4.8.4	4.7.4.8.4	X			
Shock	3.4.8.5	4.7.4.8.5	X			
Vibration	3.4.8.6	4.7.4.8.6	X			X
Durability	3.4.9	4.7.4.9	X			

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TABLE IV. Order of first article inspection. 1/

Title	Inspection	Sample 1	2	3
Defects	4.7.2	X	X	X
Insulation resistance	4.7.4.5	X	X	X
Dielectric strength	4.7.4.6	X	X	X
Rated performance	4.7.4.3	X	X	X
Weight	4.7.3	X	X	X
Voltage	4.7.4.1	X	X	X
Electromagnetic interference suppression	4.7.4.4	X	X	X
High temperature	4.7.4.8.1	X	X	
Low temperature	4.7.4.8.2	X	X	
Steam and waterjet cleaning	4.7.4.8.3	X	X	
Humidity	4.7.4.8.4	X	X	
Vibration	4.7.4.8.6	X	X	
Noise level	4.7.4.7	X	X	
Shock	4.7.4.8.5	X	X	
Durability	4.7.4.9			X

1/ First article inspection shall include an inspection of contractor records in accordance with 4.7.1.

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table V.

TABLE V. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
102	Nonconformance in design and construction (see 3.3 amd 3.3.2).	Visual
103	Faulty workmanship affecting performance (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
202	Improper finish (see 3.5).	Visual
203	Improper identification marking (see 3.6).	Visual
204	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

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4.5.2.3 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective fans and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separated from new lots and shall be clearly identified as reinspected lots.

4.5.3 Test. Each fan shall be subjected to the quality conformance test specified in table III.

4.5.3.1 Failure. Failure of any fan to pass the specified quality conformance test shall be cause for the Government to refuse acceptance of that fan.

4.6 Control tests. Control tests shall be conducted on 1 fan from each lot of 600 units consecutively produced. The fans shall be subjected to the tests specified in table III, in the order specified in table VI.

TABLE VI. Order of control testing.

Title	Inspection
Operational performance	4.7.4.2
Vibration	4.7.4.8.6
Insulation resistance	4.7.4.5
Dielectric strength	4.7.4.6
Rated performance	4.7.4.3

4.6.1 Failure. If the fan fails to pass any of the control tests, two additional fans may be selected from the lot and subjected to the control tests. Failure of either fan to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6.2 Disposition of fans. Fans which have been subjected to control testing shall not be delivered on the contract.

4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2, 3.3 and 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.7.2 Defects. Conformance to 3.3, 3.3.2, 3.5, 3.6 and 3.7 shall be determined by examination for the defects listed in table V. Examination shall be visual or by measurement with standard inspection equipment.

4.7.3 Weight. To determine conformance to 3.3.1, the fan shall be weighed.

4.7.4 Performance.

4.7.4.1 Voltage. To determine conformance to 3.4.1, attach a variable power supply to the fan and verify that it starts and operates at 11 and 30 V dc.

4.7.4.2 Operational performance. To determine conformance to 3.4.2, the rotor shall be rotated slowly and examined for rubbing and binding. Apply 26 ± 2 V dc and verify that the direction of airflow is as specified, that current does not exceed specified limit, and that there is no abnormal noise or vibration.

4.7.4.3 Rated performance. To determine conformance to 3.4.3, the fan's airflow rate shall be determined in accordance with AMCA-STD-210.

4.7.4.4 Electromagnetic interference suppression. To determine conformance to 3.4.4, the fan shall be tested in accordance with MIL-STD-462, notice 3.

4.7.4.5 Insulation resistance. To determine conformance to 3.4.5, the fan shall be tested in accordance with MIL-STD-202, method 302, condition B.

4.7.4.6 Dielectric strength. To determine conformance to 3.4.6, the fan shall be tested in accordance with MIL-STD-202, method 301. Voltage shall be maintained for not less than 1 minute.

4.7.4.7 Noise level. To determine conformance to 3.4.7, the octave band sound power levels shall be measured in accordance with accepted standard practices for air moving devices.

4.7.4.8 Environmental.

4.7.4.8.1 High temperature. To determine conformance to 3.4.8.1, the fan shall be installed in a chamber in a manner that will simulate service usage. Raise the internal chamber temperature to 160°F and maintain for a period of 48 hours while insuring the relative humidity is not in excess of 15%. Adjust the internal chamber temperature to 130°F, the highest operating temperature for which the fan is designed to operate, and maintain until temperature stabilization of the fan is reached (see 4.3.1). When raising or lowering the temperature of the chamber, the rate of temperature change shall not exceed 18°F per minute. During the highest operating temperature test phase, the fan shall be subjected to the test specified in 4.7.4.3. Return the fan to standard ambient conditions, stabilize and examine it for evidence of damage or deformation.

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4.7.4.8.2 Low temperature. To determine conformance to 3.4.8.2, the fan shall be installed in a chamber in a manner that will simulate service usage. Lower the internal chamber temperature to -70°F and maintain for a period of 24 hours after temperature stabilization. Adjust the internal chamber temperature to -65°F, the lowest temperature for which the fan is designed to operated, and maintain until temperature stabilization of the fan is reached. When raising or lowering the temperature in the chamber, the rate of temperature change shall not exceed 18°F per minute. During the lowest operating temperature test phase, the fan shall be subjected to the test specified in 4.7.4.3. Return the fan to standard ambient conditions, stabilize and examine it for evidence of deformation or damage.

4.7.4.8.3 Steam and waterjet cleaning. To determine conformance to 3.4.8.3, the fan shall be subjected to a cleaning as specified therein. Within 1 hour after completing the steam and waterjet cleaning, the fan shall be subjected to the test specified in 4.7.4.3. The fan shall be operated for 30 minutes and demonstrate no impairment of performance. The fan shall be examined for evidence of damage or deformation.

4.7.4.8.4 Humidity. To determine conformance to 3.4.8.4, the fan shall be tested according to the following procedure:

a. The fan shall be installed as in use in a chamber trap-vented to the atmosphere to prevent the build up of total pressure. No condensation shall be permitted to drip on the fan. Relative humidity shall be determined from dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the procuring activity. Air velocity flowing across the wet bulb shall be not less than 900 feet per minute (ft/min). Provision shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall not exceed 150 ft/min. Steam or distilled, demineralized, or deionized water having a hydrogen ion concentration (PH) between 6.0 and 7.2 at 73°F shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the fan by the test facility.

b. After having been dried for 24 hours at 129 ± 3°F, the fan shall be conditioned at 73 ± 3°F and 50 ± 10% relative humidity for 24 hours. Raise the internal chamber temperature to 86 ± 3°F and relative humidity to 94 ± 4%. Expose the fan to five continuous 48-hour cycles in accordance with figure 2. After completion of cycling, condition the fan for 24 hours at 73 ± 3°F and 50 ± 10% relative humidity. Accumulated moisture may be removed by turning the fan upside down or shaking. Wiping is not permitted. The fan shall be subjected to the tests specified in 4.7.4.3, 4.7.4.5, and 4.7.4.6 and examined for evidence of damage or deformation.

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4.7.4.8.5 Shock. To determine conformance to 3.4.8.5, the fan shall be mounted by normal mounting provisions to a shock machine capable of producing the shock pulse shown in figure 3. Three sawtooth shocks of 40 g and time duration of 11 ms (see figure 3) in each direction shall be applied along three mutually perpendicular axes of the fan (total of 18 shocks). After the shock test, the fan shall be subjected to the test specified in 4.7.4.2 and examined for evidence of damage or deformation.

4.7.4.8.6 Vibration. To determine conformance to 3.4.8.6, the fan shall be attached by its normal mounting means either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. The fan shall be vibrated along each axis in accordance with the test level specified in figure 1. The sweep time shall be 15 minutes for a sweep frequency range of 5 to 500 to 5 Hz. When the fan resonance, below 5 Hz are measured, the test curves shall be extended to 2 Hz and the sweep time shall be 18 minutes. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 4. The specified sweep time is that of an ascending plus a descending sweep shown and is twice the ascending sweep time shown in figure 4 for the specified range. The sinusodial cycling time shall be 30 minutes per axis. During vibration, the fan shall be operated at 27 $\pm$ 2 V dc under ambient temperature conditions. After the vibration test, the fan shall be subjected to the tests specified in 4.7.4.3, 4.7.4.5, and 4.7.4.6 and examined for evidence of damage and deformation.

4.7.4.9 Durability. To determine conformance to 3.4.9, the fan shall be subjected to 1000 hours of continuous operation under the conditions specified therein. No maintenance of any kind shall be allowed during the test except replacement of brushes after not less than 500 hours. At the completion of the durability test, the fan shall be subjected to the tests specified in 4.7.4.3, 4.7.4.5, and 4.7.4.6.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The fan is intended for use on military tracked vehicles in the M2/M3 Bradley Fighting Vehicle Systems (FVS).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article inspection is are not required (see 3.1).
c. If responsibility for inspection shall be other than as specified (see 4.1).
d. If inspection equipment shall be other than as specified (see 4.1.2).

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e. If inspection conditons shall be other than as specified (see 4.3).
f. If first article sample size shall be other than as specified (see 4.4).
g. Selection of applicable level and packaging requirments (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.1.1).

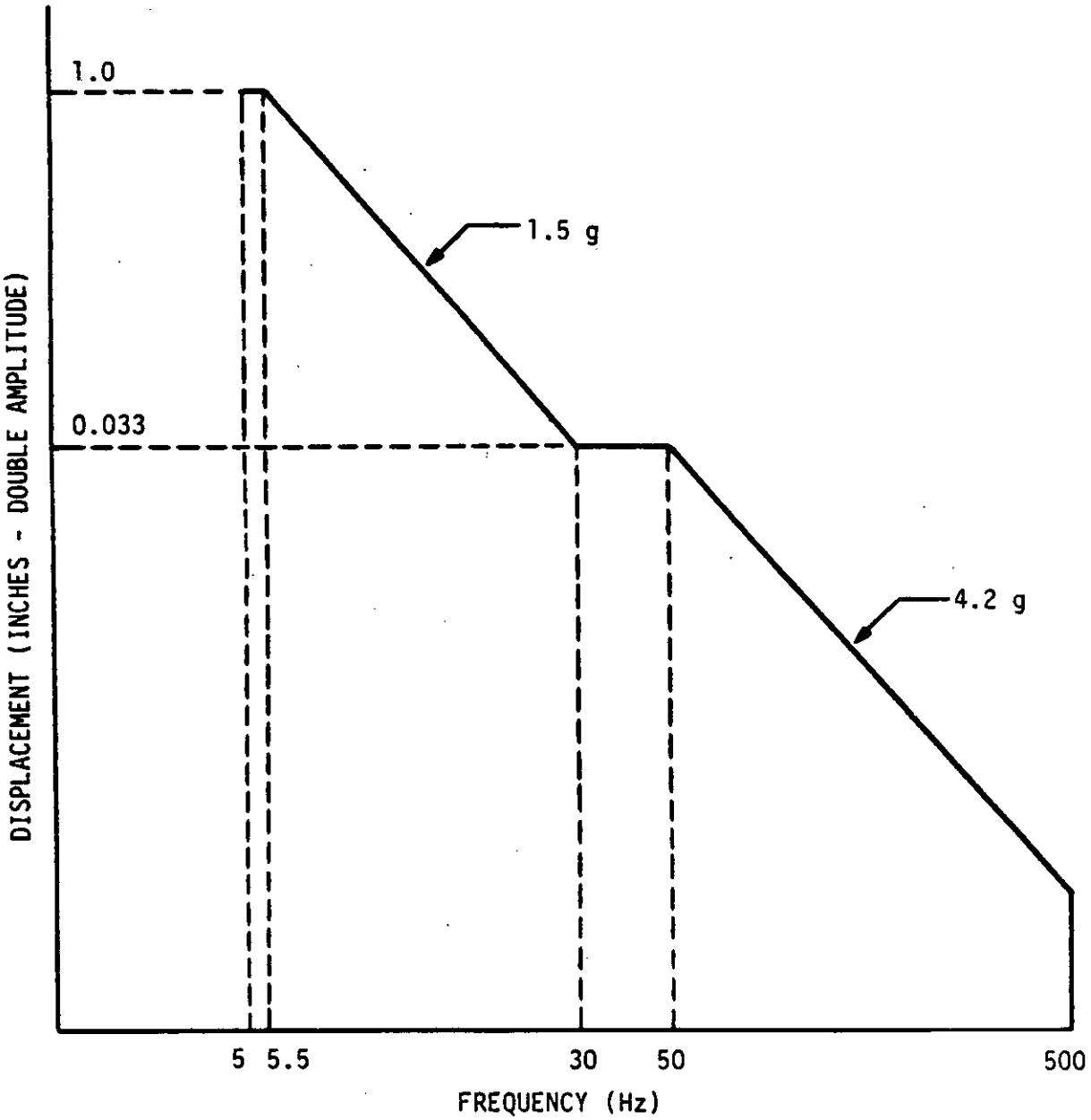
6.3.1.1 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Supersession data. This specification supersedes FMC specification 12292161, revision D, dated 24 August 1983.

6.5 Subject term (key word) listing.

Electric-powered, single-stage, fan
Fan, vaneaxial
Ventilating, fan

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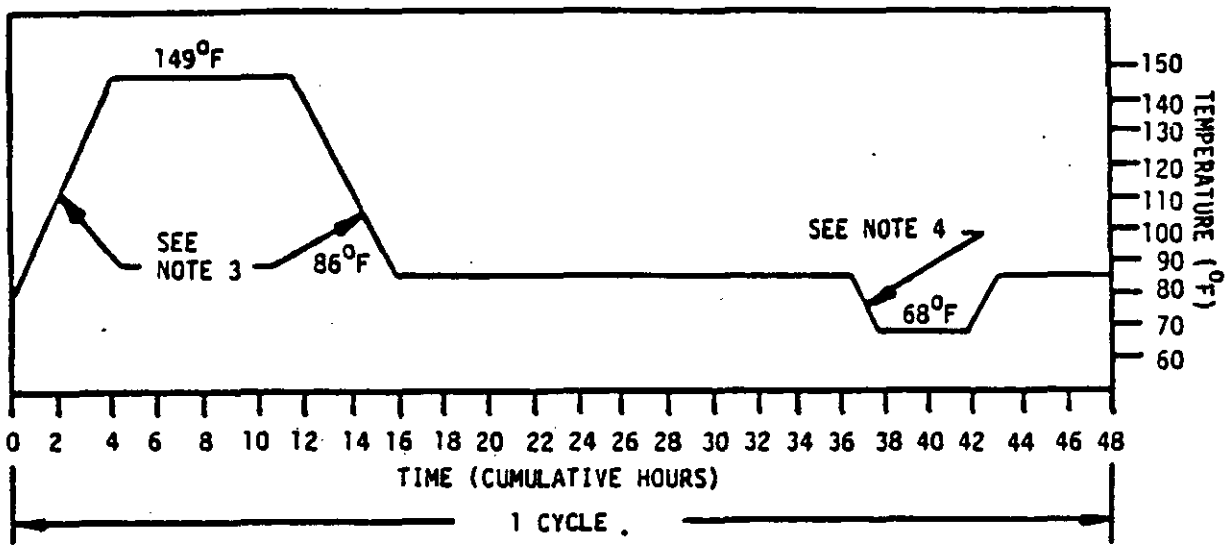


NOTES:

1. Acceleration levels: ± g (peak).
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 1. Vibration test curve.

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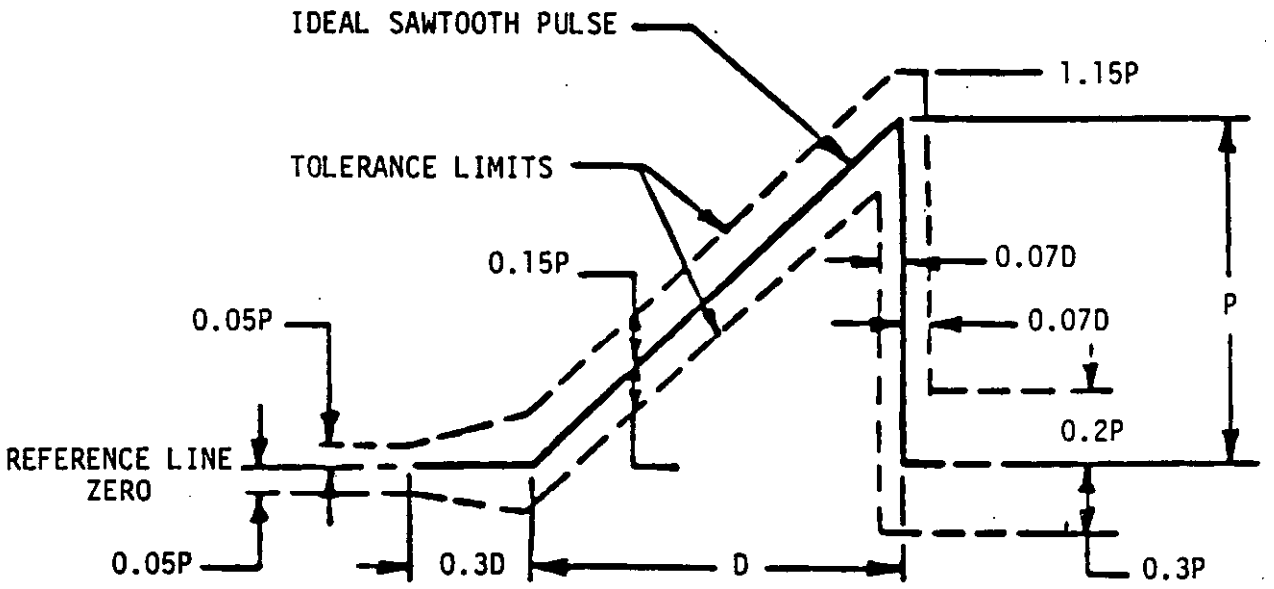


NOTES:

1. Tolerance during temperature change shall be not greater than 5°F.
2. Relative humidity shall be maintained at 94 ± 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 86 and 149°F shall be not less than 14.4°F per hour.
4. The temperature increase in this portion of the curve shall be not less than 18°F.

FIGURE 2. Humidity cycle.

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P = Peak value = 40 g
D = Nominal duration = 11 ms

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and it's duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_i \pm 0.1\ V_i$ where V_i is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 3. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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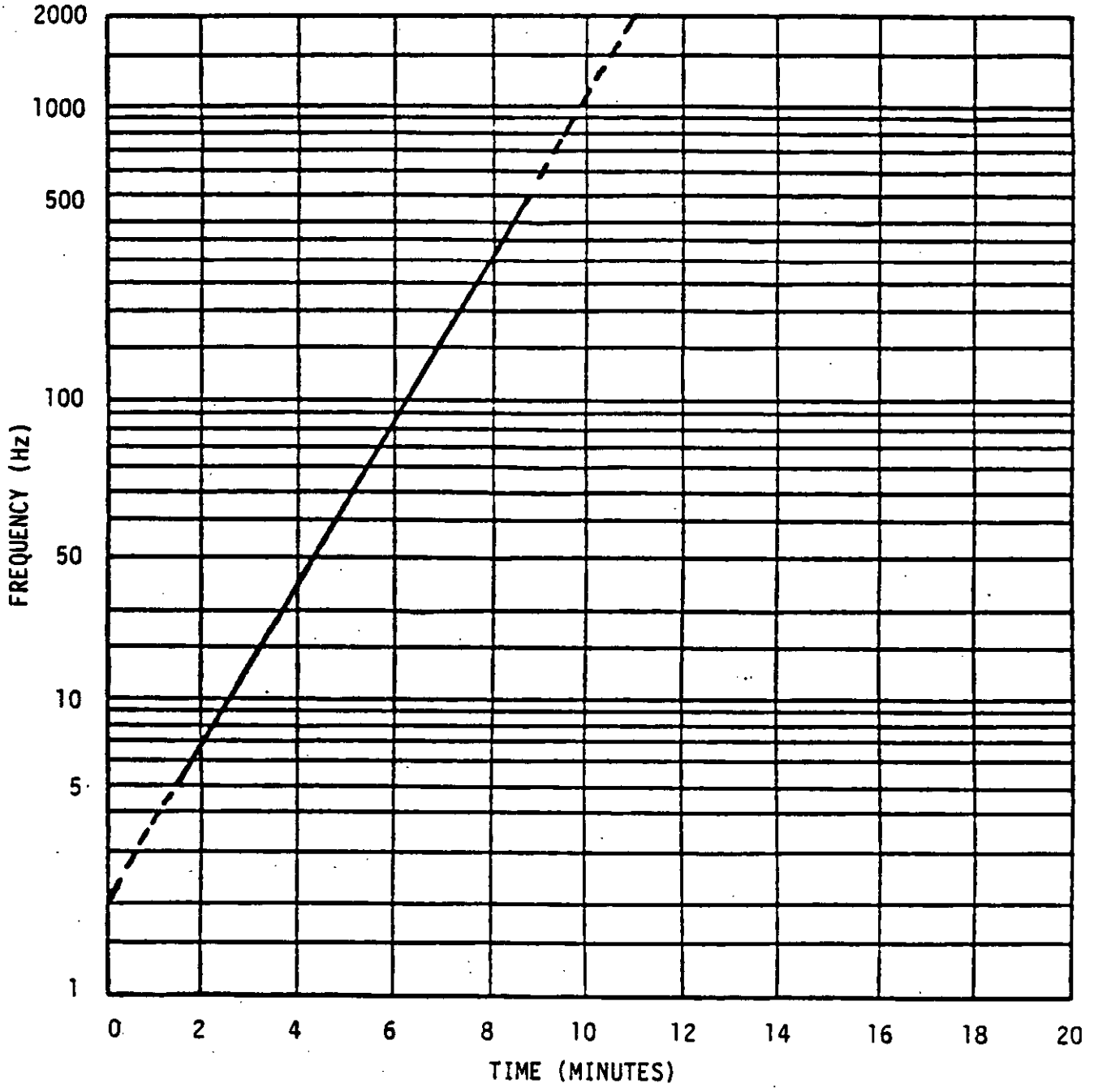


FIGURE 4. Logarithmic sweep.

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-F-62563(AT); Fan Ventilating, Vehicular Compartment

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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